

Object Detecting Augmented Reality System

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Abstract—Augmented reality technology is widely used in people's life and work, especially in visual aids. How to assist users to quickly find target objects in a complex environment is one of the problems that need to be solved in the development of augmented reality technology. To solve this problem, the paper proposes an augmented reality system based on object detection neural network and a fast method to construct specific object data set. Since it is difficult for the existing object detection network to accurately detect specific objects that are not in the data set, this paper uses the self-built data set to retrain the object detection network to ensure that the object detection network can detect specific objects. Meanwhile, we calibrate the system to ensure that the bounding box in the AR glasses output from the object detection network can accurately select the specific object. This system helps users quickly find the required items from the complex real environment and enhances the interaction efficiency between users and the real world.

Keywords—augmented reality; object detection; neural network

I. INTRODUCTION

In recent years, with the rapid development of augmented reality technology and the continuous emergence of various AR glasses, augmented reality technology has been widely used in people's daily lives. Nowadays, the mainstream augmented reality application is to use carrier tools like AR glasses to supplement the information of real objects, so as to help people better observe, analyze and understand the real world.

The real world is often diverse and complex, which makes it difficult for users to quickly select the items they need from the real environment. However, as a display device, AR glasses do not have the ability to capture and analyze the real environment.

In the field of computer vision, experts and scholars put forward the object detection technology to quickly select target objects from the complex environment. They use RGB cameras to capture images of the real environment, identify objects in the image through object detection algorithms, and then box out target objects. In recent years, with the rapid development of neural networks, excellent target detection networks such as RCNN neural network [1], YOLO neural network [2] and SSD neural network [3] have been proposed, which greatly guarantees the accuracy and real-time performance of object detection tasks.

In this paper, an augmented reality system based on object detection is proposed by combining augmented reality technology with object detection technology. Because object detection neural network trained by the official is difficult to detect specific objects that are not in the data set, this paper proposes a method to quickly construct the data set for the specific object, and trains the object detection network with the self-built data set. At the same time, we calibrate the system, making the combination of object detection network and augmented reality technology, helping users easily and quickly find and track targets from the complex real environment, so as to improve users' work efficiency. The main contributions of this paper are as follows:

- In the paper, we present an augmented reality system which can detect specific objects
- The paper presents a new data set making method, which can quickly generate effective data sets.

II. RELATED WORK

This system is mainly composed of object detection technology and augmented reality technology. In this section, this paper will introduce the related work of object detection. Then, the paper also gives a short overview of the existing calibration methods of augmented reality system.

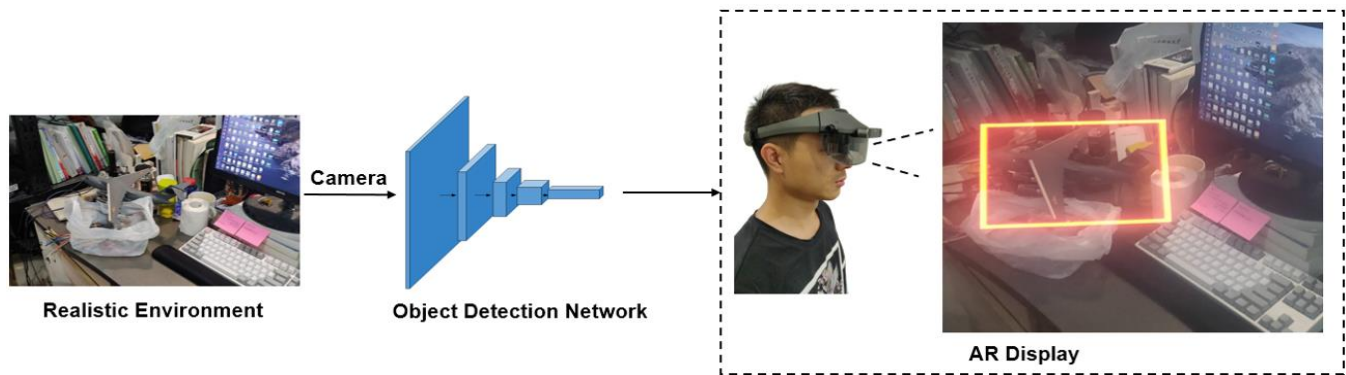


Figure 1. Main components of the system. The system uses RGB camera to capture the image of the real environment and transmit the image to the object detection network. The vertex coordinates of the bounding box are generated by the object detection network and sent to the AR display system. The AR display system renders the bounding box according to the coordinates.

A. Object Detection

Object detection is one of the hot research topics in computer vision. The traditional object detection method is usually divided into three steps. The first step is to select the region, which initially traverses the entire image by sliding the window. Since the size of the object in the image and the length-width ratio of the object cannot be determined, the scale and length-width ratio of the sliding window need to change constantly during the traversal process. The second step is feature extraction for the image. In this part, SIFT feature extraction[4] and HOG feature extraction[5] are usually used. The third step is to use SVM classifier [6] to classify the target according to the feature extracted in the second step. In recent years, some research on target classification based on characteristics has also made some progress[19][20]. However, due to the traversal operation of the sliding window, the time complexity increases and the detection speed decreases. At the same time, the diversity of objects, background and lighting environment makes it difficult to design a robust feature extraction method, which further leads to the difficulty in improving the accuracy of classification.

With the rapid development of deep learning in recent years, object detection method based on deep learning has gradually become the mainstream algorithm of object detection. The object detection methods based on deep learning model are mainly divided into two types: one-stage object detection method and two-stage object detection method.

The one-stage object detection algorithm is to directly obtain the coordinates of the target object and its category probability. This algorithm is more direct and simple than two-stage object detection algorithm. The difference between one-stage object detection algorithm and two-stage object detection algorithm is that it does not need to produce regional proposal, and the image directly generates output image through CNN network. Due to its simple network structure, the output frame rate of YOLO [2][7][8] and SSD [3][9][10] can be as high as 45 frames per second, meeting the real-time requirements. In the SSD series of algorithms, DSSD algorithm [9] uses the feature map in the process of up-sampling for prediction, while SSD algorithm [3] uses the feature map in the process of down-sampling for prediction. The feature map detected by DSSD algorithm is at the high

level of the network, and its characterization ability is stronger than that detected by SSD algorithm. FSSD algorithm[10] is different from SSD algorithm and DSSD algorithm in that it adds some feature fusion processing to make the detection more accurate.

Two-stage object detection method completes the object detection task in two steps. The first step obtains the approximate position of the target object. The second step performs a more detailed inspection based on the results in the first part. Two-stage object detection algorithm has higher detection accuracy than one-stage object detection algorithm, but the detection speed is slower. R-CNN algorithm [1] uses the Selective Search method to segment the color, texture and other information in the image to generate candidate regions. Faster R-CNN[11] uses RPN network to replace the Selective Search method, and improves the algorithm's operation speed. FPN algorithm [12] connects low-level features with high-level features, so that features of different scales have more semantic information, further generate better candidate areas, and improve the detection effect.

B. Calibration

There are two main types of display devices for augmented reality systems. One is the optical see-through head-mounted displays (OSTHMD), the other is the video see-through head-mounted displays (VSTHMD). Tuceryuan et al. described the calibration of AR system based on video see-through head-mounted displays in detail in 1995 [13]. However, the calibration of AR system based on optical see-through head-mounted displays is more complicated and changeable than that based on video see-through head-mounted displays. Early calibration of OSTHMD was manual. The manual calibration method of OSTHMD by grid proposed by Kato et al. [14] and the camera-based calibration method proposed by McGarrrity et al.[15] are both relatively complex and cumbersome. Until 2000, Tuceryuan et al. proposed the calibration method of SPAAM [16], which simplified the interaction process between the mathematical model and the calibration process. On the basis of SPAAM, scholars also proposed a series of extended calibration methods, such as SPAAM method based on video tracker [17]. In 2015, Itoh et al. calibrated OSTHMD using the optical function model of eye coordinates [18].

III. SYSTEM AND METHOD

A. System

In order to help users quickly select specific target objects from the complex real environment, this paper combines object detection network with augmented reality technology, and designs and builds an augmented reality system based on real-time target detection. This system uses RGB camera to capture the real environment and then transfers the image to the object detection network. Next, the system uses TCP communication method to transmit the result of the object detection network to the rendering engine for rendering the bounding-box. The PC transmits the rendering results to the AR display system and displays them in real time, so as to achieve the purpose of real-time detection and tracking the target object. Figure 1 shows the system structure.

1) Hardware setup

This system is mainly composed of NED+ Glass X2 series head-mounted display and a computer. NED+ Glass X2 series head-mounted display is an optical see-through AR head-mounted display. Its advantage is that it can directly observe the real scene through the head-mounted display, thus bringing users a more real AR experience. As shown in Figure 2, NED+ Glass X2 series head-mounted display is integrated with a 720P RGB camera (uncalibrated). In addition, our computer provides the computing ability with an Intel Intel Core i7-4790K CPU and an NVIDIA RTX 2080 GPU.



Figure 2. NED+ Glass X2 series head-mounted display

2) Software setup

Our system is built upon the following SDKs or software tools. This system uses NED+ Glass SDK to project the display screen of PC onto AR head-mounted display. Unity3D software provides support for AR display part of this system. OpenGL SDK based on computer graphics provides support for image rendering in AR display interface. OpenCV SDK based on computer vision provides support for basic image processing. Tensorflow deep learning framework provides a platform for the operation of object detection network.

3) Detailed system

This system preprocesses in Unity3D. This system sets the background color of Unity3D scene to black, so as to keep the perspective effect of AR system good. In order to make users find the object more clearly, the system sets the color of the bounding box to red.

This system directly uses the RGB camera integrated on NED+ Glass X2 series head-mounted display to capture complex real-time image information in the real environment. Next, the system will transmit real-time image information to the trained YOLOv3 real-time object detection network for object detection (the method will be described in the following text). In this paper, the YOLOv3 algorithm was optimized. We sent the vertex coordinates in the upper left corner and the vertex coordinates in the lower right corner of the bounding box generated by the YOLOv3 network directly to Unity3D through TCP communication. Because the image coordinate system of OpenCV is different from the image coordinate system of Unity3D, the system carries out coordinate system transformation on the vertex coordinates of the bounding box received in Unity3D. Then, the system uses OpenGL SDK in Unity3D to draw the bounding box in real time and transmit it to The AR head-mounted display for real time display.

B. Object Detection Network

1) Network selection and optimization

The selection of object detection network in this paper mainly considers two points: First, this system is an augmented reality system for real-time object detection, so the real-time performance of the system must be guaranteed. In order to achieve this goal, the system must choose a good real-time object detection network as far as possible. Secondly, in order to make the system applicable to more hardware platforms, the system needs to reduce the requirement of computing power for system operation. Therefore, on the premise of ensuring the detection effect, this paper should choose the object detection network with simple network structure as far as possible.

Based on the above two factors, the system finally chooses YOLOv3 object detection network as the basic network of the system. YOLOv3 algorithm, as one of the one-stage object detection algorithms, is much faster than object detection methods such as R-CNN. In the official document, YOLOv3 network is tested on the computer equipped with NVIDIA Titan X GPU. The delay time is only 29ms, which can fully meet the real-time demand. In addition, YOLOv3 object detection network also uses structures similar to ResNet and FPN, which also improve the object detection accuracy.

This system independently builds data sets for special objects, and uses the self-built data sets to retrain the YOLOv3 network, so as to better conduct real-time object detection for special objects in the real environment. At the same time, the system only needs to obtain the vertex coordinates of the upper left corner and the vertex coordinates of the lower right corner of the bounding box

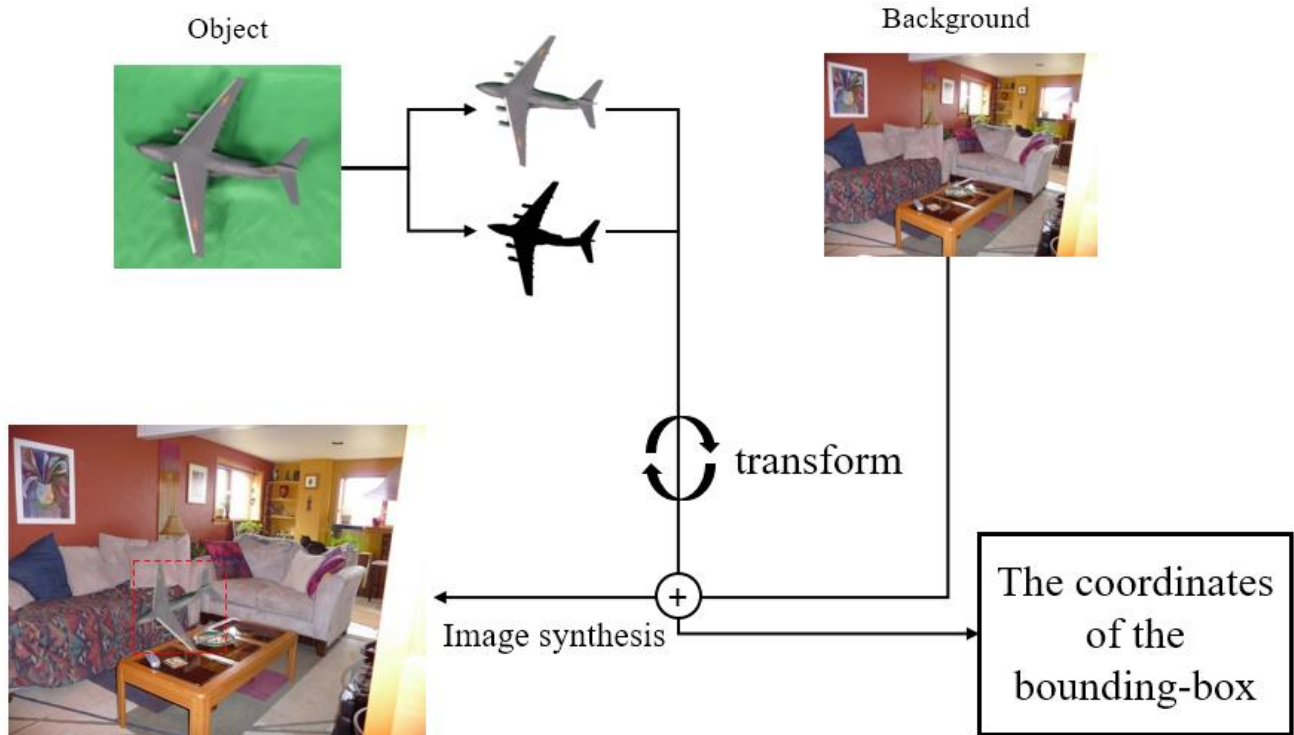


Figure 3. Data set production process . We capture the object on the green screen environment, and then extract the object's body and mask from it. We transform the object randomly and then superimpose it on the background image to obtain the data set image. At the same time, we can obtain the boundary box (the red dashed box in the figure) according to the pixel coordinates of the mask in the image.

from the output data of the YOLOv3 network, so the system simplifies the output process of the YOLOv3 network, further reducing the network delay and improving the system efficiency.

2) Data set making method

When we use neural networks for our own systems, in order for neural networks to perform better in our personalized environment, we create our own data sets. Then we use the self-built data set to train the neural network, so as to improve its stability. Therefore, how to quickly and efficiently produce data sets that meet our requirements is a crucial issue. Generally, the data set used for object detection network training is generated by manual annotation software. Marking in this way would take a lot of time. In order to solve this problem, this system proposes a method to make data sets for specific items quickly. Figure 3 shows the production process of this method.

Firstly, we set up a green screen shooting environment to capture the object from multiple angles and obtain a group of images containing only the object on the green screen background.

Secondly, this paper binarizes the green screen image to obtain the mask image of the object.

Thirdly, the mask image of the target object and its body image are transformed by random size transformation, rotation transformation and coordinate translation transformation.

Fourthly, according to the transformed mask image, we take the minimum abscissa and maximum ordinate of the pixel in the mask image as the vertex coordinates of the upper left corner of the bounding box and the maximum abscissa and minimum ordinate of the pixel in the mask image as the vertex coordinates of the lower right corner of the bounding box. Thus, the size of the bounding box of the object in the image is obtained.

Finally, a new data set is obtained by synthesizing the object image and the background image after random transformation.

C. System Calibration

In order to enable the AR display system to accurately detect and track the object, the system combines the method of SPAAM [16] to calibrate the optical see-through AR head-mounted display and RGB camera in this system. In this paper, the AR head-mounted display and the human eye are regarded as a whole and set as a virtual camera in unity environment. Then, both the virtual camera and the RGB camera are set to a fixed focal length mode with a resolution of 720P. Next, the calibration board is made in the real environment and the "X" marker is made in the center of the calibration board. At the same time, the system sets the sight device for calibration in unity environment, and sets the size and shape of the sight device consistent with the mark points in the real space. Then, the human eye looks at the mark through the AR head-mounted display and uses the mouse to



Figure 4. Example of our dataset.



Figure 5. Results of object detection of our YOLOv3 network in different environments. Moreover, the official YOLOv3 network cannot detect aeroplane models in these environments.

move the sight device so that the sight device coincides exactly with the mark. When the sight device coincides exactly with the mark point, this paper records the pixel coordinates of the mark point center in the camera image and the 3d coordinates of the sight device in Unity3D. Then, this paper changes the position of the calibration board and repeats the above operation until 12 coordinate points are recorded. Finally, this paper uses PnP algorithm to solve 12 coordinate points, so as to obtain the relative pose relationship between RGB camera and AR display, and complete the calibration.

IV. RESULT

A. Result of Dataset

According to the system's requirements, this paper only builds data sets for special objects, and trains the object detection network with the self-built data sets. This paper only takes the aeroplane model as the target object, and uses the data set production method proposed in Section 3.3.2 to produce 2482 pictures, each of which uses a completely different background environment, thus ensuring the diversity of image environment. The camera used in the production of the data set in this paper is a normal camera with a resolution of 1920×1080 and a frame rate of 25 FPS. This paper captured a total of 378 seconds of image video of the aeroplane model under the green screen background and sampled 605 different and valid aeroplane model attitude images from the video. By using image processing, we randomly transformed the attitude images of the aeroplane model mentioned above and combined with 2482 different background pictures to generate a dataset that can be used for the training of the object detection network. Figure 4 shows part of the image data in the data set, including pictures of aircraft models of different sizes and attitudes in different environments. Finally, this paper retrains the YOLOv3 neural network directly by using the self-produced data set.

B. Evaluation for Object Detection Network

TABLE I. EVALUATION RESULTS ON NETWORK

	Official	Ours
Image Size	416×416	416×416
Has the aeroplane model been detected	No	Yes
AP_{50}	-	84.45%
AP_{75}	-	44.99%
FPS	29.4	50
Time(ms)	34	20

This paper takes the aircraft model as the object to conduct a comparative test between the official YOLOv3 network and the self-training YOLOv3 network. We used a computer equipped with NVIDIA RTX 2080 GPU for the test, and the test results are shown in Table 1. The AP data in Table 1 is only for the accuracy of the object, the aeroplane model. The results showed that YOLOv3, which officials had trained, could not detect the aeroplane model. The images in Figure 5 show the object detection results of the YOLOv3 network we trained. Since the official YOLOv3 network detects more unrelated target categories than our YOLOv3 network (our YOLOv3 network only detects aeroplane models), the detection delay of the official YOLOv3 network is higher than that of our YOLOv3 network.

C. System Result

This system allows users to detect the object from many different angles. Figure 6 shows what the user sees through the AR head-mounted display. In the complex real environment, the system can still detect the target object, so as to meet the user needs. At the same time, this system can also support the user to detect and track the object in the moving process, so that users can have a more comfortable experience.

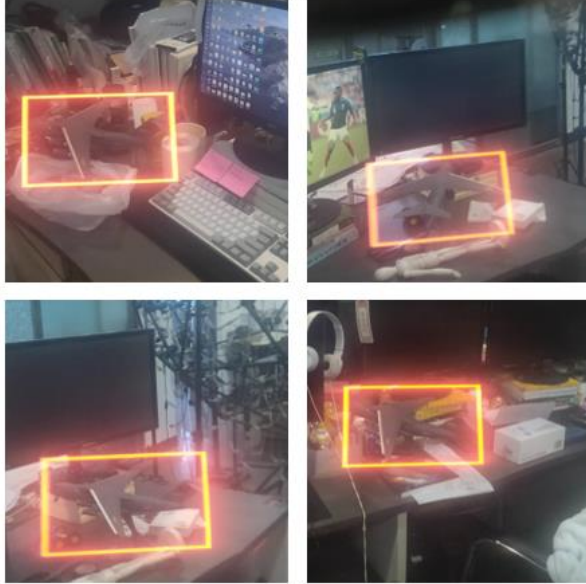


Figure 6. Display of AR system.

V. DISCUSSION

We consider the reasons why the official YOLOv3 network is not effective in detecting aeroplane models. The paper argues that the main reason is that the official data set contains only aeroplane categories and does not contain aeroplane model categories. Some people may think that the aeroplane category is not too different from the image of the aeroplane model, but the background environment of the two categories is completely different. In the real world, aeroplane usually come out of large space areas such as the sky or airports, which have relatively simple environmental backgrounds. Therefore, in the official data set of the YOLOv3 network, the environmental texture information containing the relevant images of the aeroplane is relatively simple. As a result, the YOLOv3 network, which is trained using official data sets, is difficult to detect objects such as aeroplane models in complex environment.

At the same time, this paper argues that too many object categories in the official YOLOv3 network will increase the delay time of the system and interfere with users' judgment of target objects.

VI. CONCLUSION AND FUTURE WORK

In this paper, we propose an augmented reality system based on object detection and a method for rapidly constructing data sets. In order to detect specific target

objects from the complex real environment, this paper uses the self-built data set to retrain the YOLOv3 network, and combines our YOLOv3 network with AR system display through calibration and TCP communication, so as to design an augmented reality system based on target detection technology. We believe that this system can help people quickly select the objects needed from the complex environment, to achieve the goal of improving user efficiency.

In the future, we will further optimize the object detection network, reduce the delay time of the object detection network and reduce the requirement of computing capacity of the object detection network, so as to transplant the system to mobile devices. On the other hand, we will also study the interaction mode of the system, so as to give users a better experience.

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